AppWeaver: Agentic Discovery and Composition of O-RAN Apps for AI-Native 6G

Yongchao Dang, Tao Chen, and Riku Jäntti

Abstract—Open RAN (O-RAN) opens the radio access network to a growing ecosystem of third-party rApps and xApps, allowing radio resource optimization through standardized Non-Real-Time (Non-RT) and Near-Real-Time (Near-RT) RAN Intelligent Controllers (RICs). However, building O-RAN services for emerging 6G verticals, such as connected vehicles and immersive VR/XR, remains difficult because operators must still manually discover suitable applications, understand their dependencies, configure and validate them, and coordinate them into end-to-end workflows. This article introduces AppWeaver, an agentic-AI framework that orchestrates end-to-end O-RAN workflows by using LLM-based agents to interpret operator intent, query an Agent Card registry that describes each xApp and rApp in terms of its functions, properties, and interfaces, reason about their capabilities and dependencies, and compose them into ordered, service-specific workflows. In addition, AppWeaver confines LLM agents to workflow-level orchestration over existing xApps and rApps, rather than direct RAN control, preserving a clear control-plane boundary in which native RIC applications execute low-level RAN control and monitoring actions, while agent-generated workflows are evaluated against operator policies, security constraints, validation outcomes, and rollback requirements. The resulting workflows are orchestrated with LangGraph, coordinated via Agent-to-Agent (A2A) communication, connected to xApps and rApps through Model Context Protocol (MCP) tool bindings, and evaluated on a FlexRIC testbed, a 5G-LENA ns-3 RAN and a real transport link. Over 8,712 runs on three intent sets, AppWeaver composes the correct workflow for 69–94% of intents and 55–94% of those no operator template solves, where rule-based and catalog-prompted LLM baselines score 0%; deployed, its workflows measurably change throughput, energy and delay.

Index Terms—O-RAN, agentic AI, large language models (LLMs), workflow composition.

I. Introduction

Upcoming 6G networks will evolve into programmable, artificial intelligence (AI)-native platforms for immersive, sensing, and edge-intelligence services, whose diverse requirements demand flexible and service-aware RAN automation [1]. To provide this flexibility, the O-RAN ALLIANCE[1] defines a disaggregated and programmable RAN architecture with standardized open interfaces and hierarchical RAN Intelligent Controllers (RICs), where rApps in the Non-Real-Time (Non-RT) RIC support long-term optimization and policy guidance, while xApps in the Near-Real-Time (Near-RT) RIC provide near-real-time RAN control [2]. This openness reduces vendor lock-in and enables an open ecosystem of interoperable third-party xApps and rApps for service-specific RAN automation, providing end users with more adaptive, service-aware connectivity.

Building on this openness, Hexa-X-II[2] advances intent-based management for 6G operations, where operator intents are realized through the available xApps and rApps. However, as the application catalog grows, realizing intents across a multi-vendor ecosystem raises three open problems. First, an intent must be mapped to the right xApps, rApps, APIs, and control procedures, but O-RAN defines application exposure rather than intent-driven selection and composition[3], leaving this step manual. Second, an intent may span multiple applications that depend on shared data, models, policies, and control actions, yet these dependencies are still validated in separate steps across design, deployment, and runtime. Third, an intent can trigger automated control, but conflict mitigation, authorization, lifecycle control, auditing, rollback, and human approval are not tied together into one governed execution path. These gaps make intent realization difficult to scale and govern in open, multi-vendor O-RAN systems.

To this end, this paper proposes AppWeaver, an agentic-AI framework that orchestrates end-to-end O-RAN workflows by using LLM-based agents to interpret operator intent, discover reusable xApps and rApps, reason about their capabilities and dependencies, and compose them into ordered, service-specific workflows. The main design elements and contributions of AppWeaver are summarized as follows.

- Agent Card for xApp/rApp. Each xApp and rApp is wrapped by a machine-readable Agent Card[4] that describes its functions, interfaces, control scope, resource demand, dependencies, policy constraints, and MCP tool endpoints. This allows agents to discover, compare, and reuse third-party applications without modifying their internals, while providing the seman-

Yongchao Dang and Riku Jäntti are with the Department of Information and Communications Engineering, School of Electrical Engineering, Aalto University, 02150 Espoo, Finland (e-mail: firstname.lastname@aalto.fi).

Tao Chen is with VTT Technical Research Centre of Finland, 02150 Espoo, Finland (e-mail: tao.chen@vtt.fi).

[1]https://www.o-ran.org/

[2]https://hexa-x-ii.eu/

[3]https://www.o-ran.org/specifications

[4]https://www.agentcard.net/

TABLE I: Classification of representative agentic O-RAN systems by capability.

System	Control paradigm	Intent	Reuse & composition		Conflict & assurance			Evaluation
			Disc.	Comp.	Confl.	Valid.	Adapt.	
RIC-O [2]	Optimization-based RIC placement	✗	✗	✗	✗	✗	∼	Kubernetes
AutoRAN [3]	LLM-driven deployment automation	✓	✗	∼	✗	✓	∼	OAI/srsRAN/ARC
OREO [4]	Optimization-based xApp deployment	✗	∼	∼	∼	✗	✗	Colosseum emulator
RAN assurance [5]	Multi-agent assurance	∼	✗	∼	✓	✓	✓	Simulation
AGORAN [6]	Multi-agent intent negotiation	✓	∼	✗	✓	✗	∼	OAI/FlexRIC
AgentRAN [7]	Hierarchical multi-agent control	✓	✗	∼	∼	∼	✓	OAI (OTA)
Multi-scale [8]	Multi-timescale multi-agent control	✓	✗	✗	∼	✗	✓	Architecture
Multi-agentic rApp [9]	Multi-agent policy control	✓	✗	∼	✓	∼	∼	Simulation
MX-AI [10]	Multi-agent observability and control	✓	✗	∼	✗	∼	∼	OAI/FlexRIC
AppWeaver (ours)	Multi-agent workflow orchestration	✓	✓	✓	✓	✓	∼	FlexRIC/ns-3

Control paradigm: control approach adopted. Intent: natural-language interface. Disc.: discovery of reusable xApps/rApps. Comp.: composition into executable workflows. Confl.: cross-application conflict handling. Valid.: pre-deployment sandbox or twin validation. Adapt.: runtime adaptation and rollback. Marks: ✓ demonstrated or core focus; ∼ partial or by-design; ✗ not addressed.

tic grounding needed for dependency reasoning and compatibility checking.

- Agentic workflow composition. Based on the Agent Card abstraction and the Agentic Resource Discovery (ARD) principle[5], LLM agents translate operator intent into workflows by discovering suitable xApps and rApps, reasoning about their capabilities and dependencies, and coordinating execution through A2A communication and a LangGraph-based state machine. This enables reusable O-RAN applications to be selected, connected, and orchestrated for different service requirements without manual workflow construction.
- Agentic workflow testing and deployment. Each generated workflow is tested before deployment through interface compatibility checking, policy verification, rollback planning, and human-in-the-loop approval when required. Only validated and approved workflows are deployed through standardized O-RAN interfaces, while operator policies, auditing, and rollback mechanisms keep execution observable, controlled, and reversible.

These agentic capabilities enable an intent-to-deployment pipeline that covers application discovery, workflow composition, predeployment validation, and governed execution across reusable O-RAN applications. The remainder of this paper is organized as follows. Section II reviews related work on agentic AI for O-RAN. Section III introduces the AppWeaver architecture. Section IV presents the prototype and its evaluation on FlexRIC, ns-3 and a real transport link. Section V concludes the paper and outlines future directions.

II. Agentic AI for O-RAN

Agentic AI for O-RAN refers to governed LLM-based agents that interpret operator intent, retrieve external knowledge, reason over network context, plan actions, and invoke O-RAN APIs and tools for automation. Unlike traditional rule-based network agents that follow predefined policies and fixed control logic, these agents interpret natural-language instructions and adapt their behavior to dynamic network conditions. Recent work explores this direction from three complementary perspectives, namely xApp and rApp development, O-RAN component deployment and lifecycle management, and intent-driven network automation, summarized in Table I.

A. Agentic AI for xApp and rApp Development

Developing xApps and rApps requires a heavy engineering burden, as each application requires RAN control logic, E2 and A1 service model integration, RIC SDK interfacing, and validation on the target platform. Agentic AI is being leveraged to reduce this burden, with LLM-based agents translating natural-language requirements into application code, service model bindings, configuration artifacts, and test templates, and supporting adaptive reasoning inside xApps for radio resource management [11]. However, these efforts mainly target the development of individual applications and treat the LLM as a code-generation or reasoning engine, while agentic capabilities such as tool invocation, knowledge retrieval, multi-step planning, and multi-agent collaboration remain underexplored.

B. Agentic AI for O-RAN Deployment and Management

O-RAN deployment and management involve placing disaggregated functions across cloud-native infrastructure, allocating resources, resolving inter-application conflicts, and verifying behavior before execution. To automate these tasks, existing studies combine optimization, learning-based prediction, and agentic decision-making. RIC-O optimizes the placement of distributed near-RT RIC instances across the cloud-edge continuum through

[5]https://agenticresourcediscovery.org/

dynamic clustering of radio nodes [2], while graph neural network based methods predict conflicts among xApps before deployment [12]. AutoRAN translates high-level operator intent into deployment, configuration, and testing artifacts for multi-vendor O-RAN systems [3], whereas OREO formulates xApp deployment as a joint optimization problem over application selection, function sharing, configuration, and resource allocation [4]. Assurance-oriented frameworks add explicit verification stages to validate AI-generated decisions before execution [5]. Although these efforts reduce the operational complexity of deploying and managing O-RAN components, they mainly assume that the candidate applications are already known and focus on their placement, configuration, or validation, rather than on discovering reusable xApps and rApps and assembling them into service-specific workflows.

C. Agentic AI for Intent-Based RAN Automation

Intent-based RAN automation enables operators to specify high-level service objectives in natural language, which collaborating agents translate into control actions; a key challenge is how such intents are interpreted and resolved in multi-stakeholder settings. AGORAN addresses this challenge through intent negotiation, where LLM-based broker agents reconcile conflicting requirements under governance constraints and translate the agreed outcomes into enforceable policies through standard O-RAN interfaces [6]. Building on this direction, subsequent work aligns agent structures and computational models with O-RAN's hierarchical, multi-timescale control loops, as in AgentRAN, which coordinates rApp, xApp, and dApp agents through A2A communication and MCP-based tool invocation [7], and multi-scale frameworks that map LLMs, small language models (SLMs), and wireless foundation models to the Non-RT, Near-RT, and real-time loops to meet heterogeneous latency requirements [8]. The direction further extends to concrete implementations and system-level demonstrations, including conflict-aware policy generation through collaborative rApp agents [9], [13] and live prototypes such as MX-AI, which provides end-to-end observability and closed-loop control in operational deployments [10].

Despite advances in application development, deployment automation, intent understanding, architectural design, and closed-loop execution, existing systems commonly assume that the applications required to realize a service are already known. However, an end-to-end service often requires coordinating multiple reusable applications rather than executing a single xApp or rApp in isolation. For example, traffic steering typically involves KPI monitoring, anomaly detection, QoE prediction, traffic-steering optimization, and RAN control [14]. Current approaches therefore map operator intent to actions over predefined xApps and rApps, but do not address how such applications are discovered in an open, multi-vendor ecosystem, verified for interface and control-scope compatibility, and composed into validated executable workflows. As O-RAN application ecosystems expand and services increasingly span multiple vendors, this gap in application discovery and composition becomes a key barrier to fully autonomous service orchestration, motivating the design of AppWeaver.

III. AppWeaver Architecture and Workflow

A. AppWeaver Architecture

Figure 1 shows the AppWeaver architecture, which sits between the operator specifying the intent and the RAN vendors providing the RIC and its xApps and rApps. AppWeaver does not replace existing O-RAN applications but orchestrates them into service-specific workflows through discovery, composition, validation, and governed execution. At its core, a LangGraph-based orchestration layer[6] runs a Planner agent that interprets the operator intent and coordinates four orchestration agents over the A2A bus[7]. The Discover agent identifies candidate xApps and rApps, the Compose agent assembles them into an ordered workflow, the Validate agent checks the workflow before deployment, and the Deploy agent applies the validated workflow to the RIC. These agents draw on two shared resources, an Agent Registry that stores the Agent Card and endpoint of each available xApp and rApp for capability-based selection, and a shared memory that holds the model state, running summaries, and workflow context across reasoning steps. Discovery is grounded in the registry by BM25 retrieval over the Agent Cards, and a feedback loop lets deployment outcomes refine later decisions.

AppWeaver operates across two environments, the live RAN and a network digital twin. On the live RAN, a per-application domain agent wraps each selected xApp or rApp and reaches it through an MCP client[8] connected to the MCP server co-located with the application in the Near-RT or Non-RT RIC. The server exposes monitoring and control functions as callable tools, so the domain agent orchestrates the workflow while the application executes the E2 and A1 control actions. In parallel, the Validate agent connects to a digital twin at two levels: an E2 twin (a second RIC with an emulated E2 node) on which every E2 step is dry-run to catch interface, lease and acknowledgement errors, and a network twin (a RAN simulator) that predicts the workflow's KPI impact. Only workflows that pass are applied to production. This separation gives AppWeaver a safety boundary by testing workflows before they reach the live RAN, while keeping all E2 and A1 control actions within existing xApps and rApps and their native control loops.

[6]https://www.langchain.com/langgraph
[7]https://a2a-protocol.org/latest/specification/
[8]https://modelcontextprotocol.io/specification/

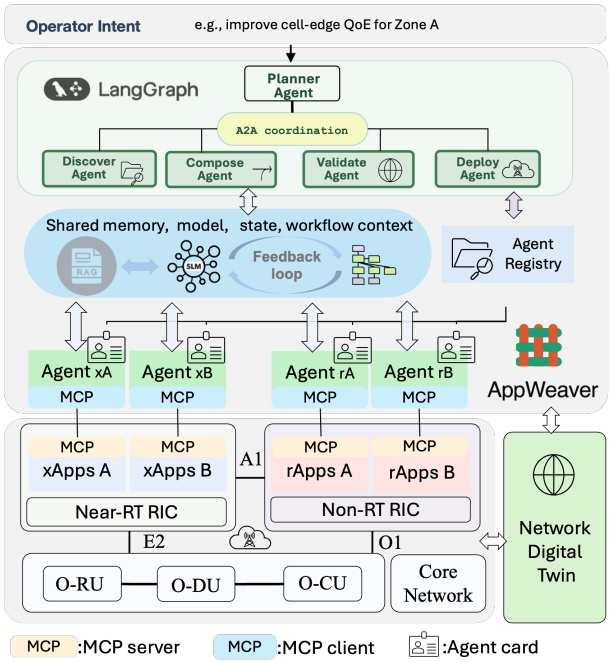


Fig. 1: AppWeaver architecture.

B. AppWeaver Workflow

The AppWeaver workflow moves through four stages, discovery, composition, validation, and deployment, all driven by the LLM-based agents in the AppWeaver core; Fig. 2 shows them for a QoE-steering intent, taken from a recorded run. Given an operator intent, the Planner agent interprets the request and the objective it implies. The Discover agent then retrieves candidate applications from the Agent Registry by BM25 search over the Agent Cards, matching the intent to advertised capabilities rather than names; the intent in Fig. 2 names no application, and the 19 retrieved cards include a QoE-prediction rApp, a traffic-steering rApp, a UE QoS-control rApp and their deprecated releases.

intent	"Steer at-risk video users away from the congested cell before their QoE drops." (no skill or app named)
discover	BM25 over 80 Agent Cards, k=8 → 19 candidates in 5 ms, incl. qoe_prediction.rApp, traffic_steering.rApp, ue_qos_control.rApp and the deprecated v0 variants.
compose	Planner (gpt-oss-120b, 31.0 s): KPM → qoe_prediction → traffic_steering → RC; v0 variants rejected.
check	Card checks: dependency, interface, control scope, timescale, lifecycle → 0 errors, 2 warnings (second RC writer serialised; near-RT step after non-RT step).
validate	Twin dry-run of the RC step over E2 on the twin RIC: acknowledged in 7.2 s; no busy lease → deployable at 38.2 s.
deploy	Four calls in chain order (KPM observe, QoE predict, steering policy, RC control): 4/4 acknowledged, lease wait 0 s.
baselines	Rule router and catalog LLM: KPM → SLICE → TC → RC (wrong, 3/3 repeats); AppWeaver correct 3/3.

Fig. 2: AppWeaver trace for a QoE-steering intent, from a recorded run (Batch B, 81-card registry).

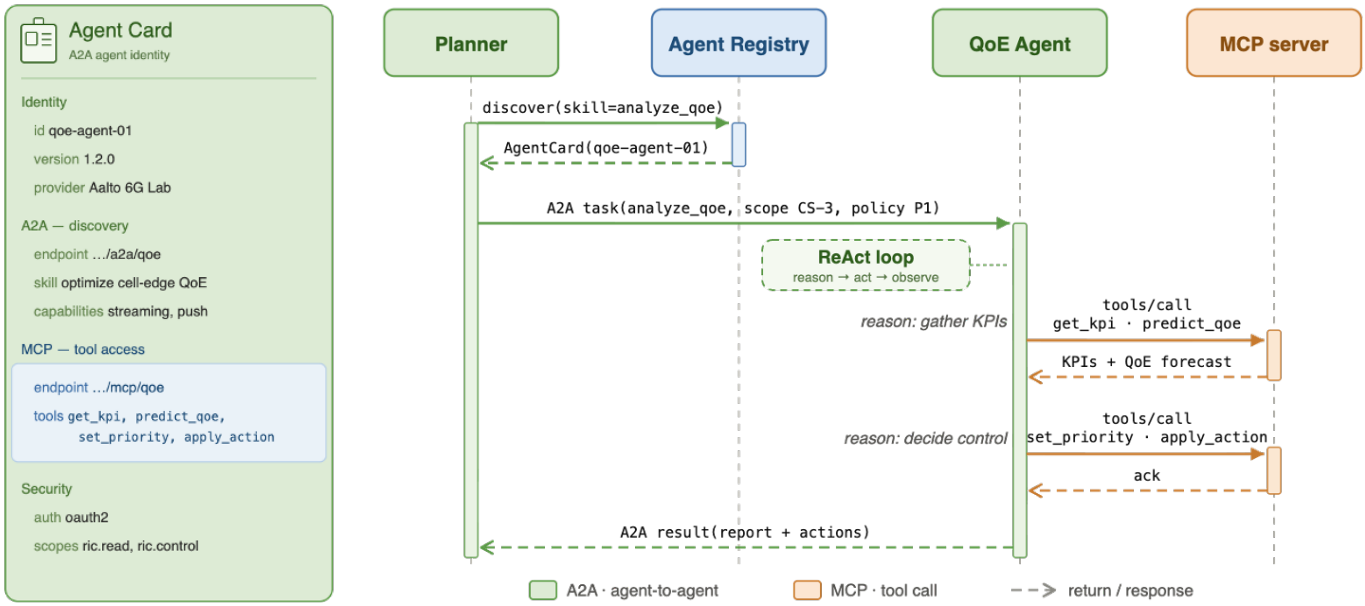
The Compose agent turns these candidates, their tool schemas, and the operator constraints into an ordered workflow: here KPM monitoring, then QoE prediction, then traffic steering, then RAN control, rejecting the deprecated releases. Before anything runs, the chain is checked against the Agent Cards for dependencies (each step's inputs are produced upstream), interfaces, control scope (no two writers on one control resource without serialization), timescale and lifecycle; errors return the plan to the Planner for repair, warnings are logged. The Validate agent then dry-runs each E2 step on a twin RIC and checks that no other workflow holds the applications' leases, and the Deploy agent executes the steps in order, each an MCP tool call whose result, including the application's E2 control acknowledgement, is written back into the shared state. KPI prediction on a network twin is evaluated in Section IV.

C. Agent Interfaces and Procedure

Figure 3 details the AppWeaver interfaces for exposing, discovering, and invoking third-party xApps and rApps, using the QoE-steering intent of Fig. 2 as an example. Fig. 3(a) shows the Agent Card, a machine-readable JSON descriptor registered in the Agent Registry that lets AppWeaver treat each xApp or rApp as a discoverable and reusable component. It specifies the application's capabilities, such as KPI monitoring, QoE prediction, mobility optimization, or QoS control; supported interfaces and data models, including E2SM-KPM, E2SM-RC, and A1 policy types; control scope over cells, slices, UEs, and tunable parameters; and resource requirements, operational constraints, and MCP tool endpoints used to monitor, configure, and control the RAN through the underlying application. Because this description is maintained outside the application, AppWeaver translates operator intent into RAN control through existing xApps and rApps without modifying their internal logic or driving O-RAN interfaces directly. A new application is onboarded by registering its Agent Card.

The MCP steps of Fig. 3(b) show how a domain agent invokes application tools through an MCP client connected to the MCP server co-located with its xApp or rApp. The agent first sends a tools/list request to discover available monitoring and control tools, then issues tools/call requests to invoke selected tools, such as get_kpi and predict_qoe for observation, and set_priority and apply_action for control. The MCP server translates these calls to the underlying application, and the xApp or rApp executes RAN operation over E2 or A1. This keeps the agent at the orchestration level, hides vendor-specific management APIs behind a uniform tool interface, and makes each external operation, whether telemetry collection, invocation, or policy update, an explicit typed action that can be logged, validated, and replayed.

Using Agent Cards and MCP tools, the orchestration agents search the Agent Registry, select suitable domain



(a) Agent Card

(b) Discovery and invocation sequence (A2A and MCP)

Fig. 3: AppWeaver agent interfaces and procedure.

agents, and assign tasks over the A2A coordination plane, as shown by the A2A steps of Fig. 3(b). For the QoE-steering intent, the Discover agent queries the registry and receives matching Agent Cards, including the QoE-prediction rApp and its endpoints. The Planner agent then assigns the corresponding domain agent an A2A task carrying the objective, scope, and policy constraints. The domain agent calls the application's MCP tools, here the QoE forecast, and returns an A2A result with the report to the Planner agent. This separation turns a high-level intent into concrete, traceable tool invocations without orchestration agents directly touching O-RAN interfaces.

The Agent Card and MCP abstractions expose heterogeneous xApps and rApps as discoverable, reusable agents, and the ReAct loop composes them through iterative reasoning, tool invocation, and observation into a validated workflow. AppWeaver serves as an intent-translation layer between the operator and vendor-provided RIC applications, mapping high-level objectives onto existing xApps and rApps through their agent interfaces. This layering separates timescales, with AppWeaver orchestrating at seconds to minutes while the applications retain native E2 and A1 control in their own loops, keeping orchestration separate from control execution.

IV. AppWeaver Evaluation

A. Prototype Setup

The AppWeaver LangGraph orchestrator runs next to a FlexRIC Near-RT RIC on one host; the Planner uses gpt-oss-120b[9] served by vLLM on an NVIDIA DGX Spark (GB10, 128 GB unified memory), with reasoning effort medium, temperature 0.2 and a fixed seed. The real RIC applications are four FlexRIC xApps used over E2, KPM monitoring (E2SM-KPM), slice control (SLICE), transport control (TC) and RAN control (E2SM-RC), plus MAC, RLC, PDCP and GTP monitors; together they form the 9-card paper registry. The rApps (QoE prediction, traffic steering, energy saving and 29 other families, each with vendor variants, deprecated releases and distractors) are emulated decision services behind standard tool interfaces; with them the registry grows to 81 Agent Cards. The E2 node used for the composition benchmark is FlexRIC's emulator, whose KPM values are synthetic; the Validate agent dry-runs each E2 step on a second, independent RIC and emulator (the E2 twin), which checks interfaces, leases and acknowledgements. Network effect is therefore measured separately, on two substrates with real dynamics: a 5G-LENA ns-3 RAN (four gNBs at 3.5 GHz) attached to FlexRIC over E2SM-KPM/RC, where a second ns-3 instance with a different seed acts as the KPI twin, and a real Linux transport bottleneck for TC, driven by an E2 agent whose TC service model applies the received queue control to the link. All code, intents, Agent Cards and run records are public[10].

B. Benchmark and Methods

Three intent sets are used. Batch A has 50 operator intents over the paper registry: 11 match one of the three operator workflow templates (known) and 39 do not (open-world), 27 of which no template solves even partially.

[9]https://huggingface.co/openai/gpt-oss-120b

[10]https://github.com/Dangy1/flexric-py310/tree/appweaver-revision

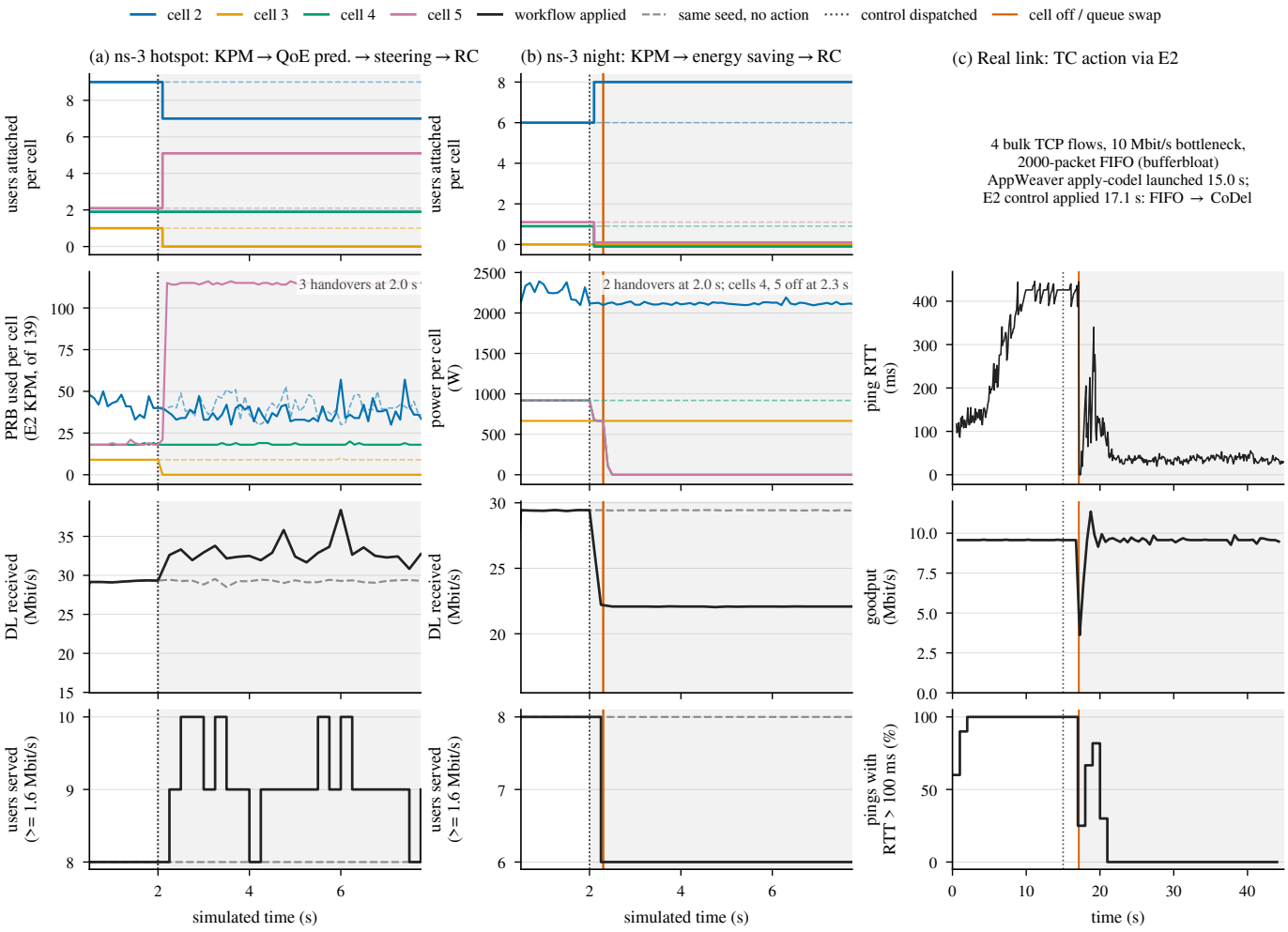


Fig. 4: System KPIs while a composed workflow acts (live instance; dashed: same seed without the action; shaded: after the action). Rows: users attached per cell and cell load or power from the E2 KPM stream, received downlink throughput, and users served (at least half their offered rate). (a) ns-3 hotspot, KPM→QoE prediction→traffic steering→RC, seed 104; (b) ns-3 night load, KPM→energy saving→RC; (c) real 10 Mbit/s link, AppWeaver's TC action (FIFO→CoDel) sent over E2: RTT, goodput and share of pings above 100 ms.

Batch B has 109 intents over the 81-card registry and adds 59 unseen-capability intents that need an rApp absent from every template; Discover retrieves the top 8 cards by BM25, and an ablation puts all 81 in the prompt. Batch C has 60 held-out intents written after the design was frozen (30 known, 30 open-world). Each intent is labeled with the applications it requires and their partial order; a composed workflow is correct if it contains every required application in a valid order and any extra step is read-only (rule accuracy, defined before the runs). Six intents in A and B hold a seeded conflict, an application kept busy by another workflow. AppWeaver is compared with a rule router (keyword templates), a single-app router, an LLM without the catalog, the same LLM with the full catalog in its prompt (single pass, no Agent Card checks), and AppWeaver without repair. Each intent runs 5 (A) or 3 (B, C) times, 8,712 records in total, with planner caches off.

C. Workflow Composition

Fig. 5(a) shows that AppWeaver composes the correct workflow for 70.4% of the Batch A intents, 68.8% in Batch B and 93.9% in the held-out Batch C, against 34.8%, 16.8% and 53.3% for the catalog-aware LLM and 36.0%, 16.5% and 20.0% for the rule router. The baselines score well only where a template fits: on the intents no template solves, both score 0% in every batch, while AppWeaver reaches 67.4%, 55.6% and 93.9% (Fig. 5(b)); on Batch B's unseen-capability intents it reaches 82.5%. Fig. 2 shows one such case: both baselines return the generic KPM→SLICE→TC→RC template, while AppWeaver composes the QoE-prediction and steering rApps. Retrieving 8 of 81 cards gives the same accuracy as showing all of them (68.8% both) with about one third of

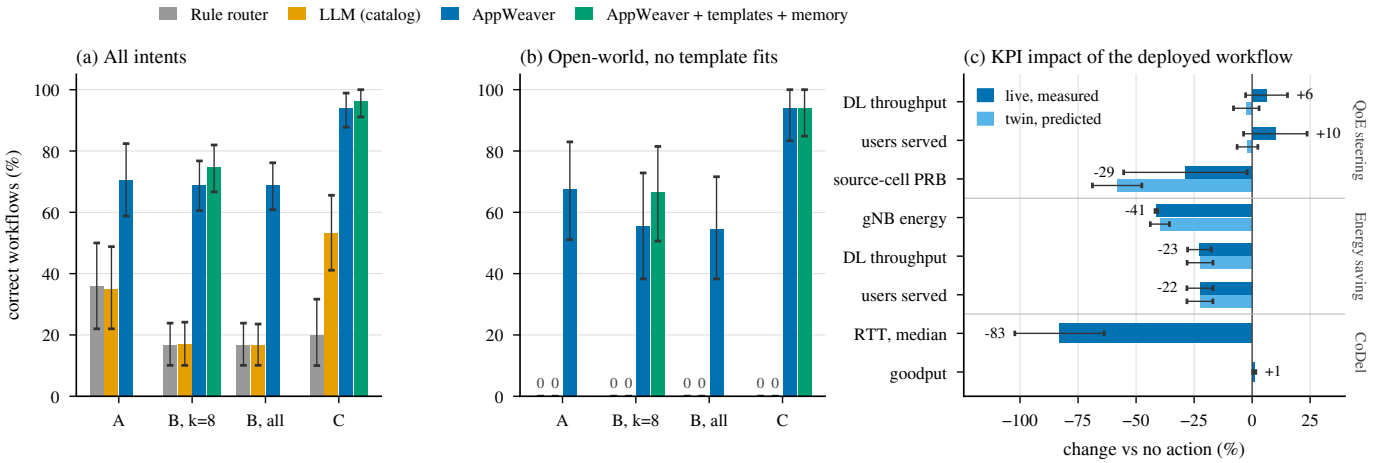


Fig. 5: (a) Rule accuracy per batch and (b) on open-world intents that no operator template solves, 95% intent-clustered bootstrap CIs; (c) KPI change caused by the deployed workflow versus the same instance without it, live measurement and twin prediction, mean ± sd over 5 seed pairs (ns-3) or 5 repeats (real link, no twin).

the prompt tokens (9k versus 25k) and a median planning time of 32.5 s instead of 36.2 s. Adding the operator templates and an experience memory of earlier plans as retrieved examples raises accuracy to 74.6% (B) and 96.1% (C). A second planner, GLM-5.3-Flash, gives the same accuracy on Batch C (93.9%) and the same 0% for the template-free baselines, so the gain is not specific to one model.

D. Network Effect of the Composed Workflows

Fig. 4 follows the system KPIs while a composed workflow acts, and Fig. 5(c) summarizes the change it causes over five seeds. On the ns-3 hotspot, the QoE-steering chain hands users from the loaded cell to a neighbor at $t = 2$ s; the source cell's PRB use falls by 29% and the neighbor's rises within 0.3 s. Users served rise by 10% and received throughput by 6.2% on average: in this layout the simulator leaves some hotspot users unserved without any action, and steering restores service to two of them in two of the five seeds, while load moves without a throughput change in the others. The baselines' workflow for this intent (KPM→SLICE→TC→RC) contains no steering step and cannot produce this effect. In the night scenario, the energy-saving chain moves two users and switches two cells off: gNB energy falls by 41.3%, but those two users lose service, so users served and throughput fall by 22–23%. The ns-3 twin, which knows the layout but not the live mobility and fading, predicts all three energy-saving effects within 3 percentage points (correct sign in 5/5 seeds), and the steering load shift in 5/5 seeds, but it misses the throughput gain from restored users. The twin thus carries the information needed to stop the harmful energy policy, provided the acceptance rule tests users served or throughput: a rule on energy and load alone accepts it in all five seeds. On the real link (Fig. 4(c)), the TC action that AppWeaver deploys reaches the queue 2.1 s after launch; under four bulk TCP flows it cuts the median RTT from about 420 ms to 35 ms and the share of pings above 100 ms from 100% to 0%, with unchanged goodput (9.6 Mbit/s), in all five repeats.

E. Reliability, Conflicts and Latency

Across all batches, 97–100% of AppWeaver's control calls are acknowledged by the target xApp, and no plan references an application outside the registry, whereas the LLM without the catalog produces 705 such references in Batch A. On seeded conflicts AppWeaver never preempts the busy application: it waits for the lease and completes the intent in 37% (A) to 44% (B) of cases, with no unresolved refusal, while the rule router runs into the lease and completes with the wrong workflow in 83% of its cases. The median time from intent to deployable workflow is 26–36 s, dominated by the planner; the E2 dry-run adds about 8 s. This is slower than the single-pass LLM (7–9 s) but within the Non-RT RIC timescale (>1 s)[11].

V. Conclusion and Future Work

This article presented AppWeaver, an agentic-AI framework that translates operator intent into validated O-RAN workflows by discovering reusable xApps and rApps, composing them, and deploying them under operator governance. AppWeaver wraps each application in a machine-readable Agent Card for capability-based discovery, coordinates Planner, Discover, Compose, Validate, and Deploy agents over A2A within a LangGraph orchestrator, binds applications through MCP tool interfaces, and rehearses every candidate workflow on a network digital twin before it reaches the live RAN. By confining LLM agents to workflow-level orchestration, AppWeaver keeps all E2 and A1 control actions inside the native xApps, rApps, and

[11] O-RAN.WG1.OAD, https://specifications.o-ran.org/.

RIC control loops, so reasoning and coordination sit above the RAN while execution remains within standardized interfaces under operator policy, auditing, and rollback. The evaluation supports this design. Over three intent sets and registries of up to 81 Agent Cards, AppWeaver composes the correct workflow for 69–94% of intents and for 55–94% of those no operator template solves, where the baselines score 0%, with grounded, acknowledged control calls and no pre-emption of busy applications, at a latency within the Non-RT timescale. Deployed on a 5G RAN model and a real transport link, the composed workflows change throughput, energy and delay measurably, and the network twin predicts these effects closely enough to stop a harmful policy, provided its acceptance rule tests the KPIs that policy hurts.

Several directions remain for future work. First, runtime adaptation is currently partial, and tighter closed-loop self-healing together with learned digital-twin surrogates would strengthen the observe-and-adapt stage and reduce validation cost. Second, discovery is performed over a curated Agent Card catalog, and scaling to large multi-vendor repositories and open marketplaces raises trust, versioning, and conflict-resolution challenges. Third, rApps are emulated as test endpoints in the current testbed, and deploying AppWeaver against production-scale Non-RT RIC applications in collaboration with industry partners is a natural next step. Finally, the A2A and MCP bindings used here are pragmatic mechanisms, and aligning them with the RAN-grade requirements on semantic contracts, authorization, behavioral bounds, traceability, and rollback now being studied in O-RAN working groups and 3GPP SA5 is important work toward a standardized agent-to-O-RAN interaction model.

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